



Aror University of Art, Architecture, Design & Heritage Sukkur.

BS(Artificial Intelligence) Fall-2023 Robotics

Course Title: Robotics
Course Code:
Credit Hours: (2+1)
Course Instructor: Bilal Ahmed
Electronic mail: bilal.faculty@aror.edu.pk

Description:

This course introduces students to the fundamental concepts, tools, and practices of robotics, spanning embedded systems, sensors and actuators, and modern robotic software frameworks. Students begin with hands-on microcontroller programming (Arduino, ESP32) and sensor/actuator interfacing, progress through IoT-based communication, and culminate in robot modeling, kinematics, and simulation using ROS 2 and Gazebo. The course emphasizes practical, project-based learning, with each theoretical concept reinforced through a corresponding lab exercise, ending in a final integrated robotics project.

Aims and Objectives:

1. Explain fundamental concepts, history, and applications of robotics across industries, including the role of AI integration in modern robotic systems.
2. Understand robot kinematics, sensors, actuators, and control systems for robotic motion and interaction.
3. Learn robot programming solutions using platforms such as ROS, Python, or C++, applied through Arduino/ESP32 embedded development and ROS 2.
4. Demonstrate practical skills in robot assembly, programming, and testing through hands-on projects, culminating in an integrated final robotics project.



Aror University of Art, Architecture, Design & Heritage Sukkur.

Assessment:

S. No	Assessment Activities	Percentage	Total Activities
1.	Sessional: Quizzes/ Assignments (Quizzes & Assignments)	30%	Multiple
2.	Mid Term Exam	30%	1
3.	Final Exam	40%	1

Course Learning Outcomes (CLOs):

No.	Course Learning Outcome	Domain	GA	Assessment Tool
C1	Understand basic robotic problem-solving steps and control logic, including sensor-driven decision making and conditional/loop-based control constructs.	C2	GA2	Class Participation, Quizzes, Mid Exams., Assignments, Final Exam
C2	Demonstrate basic embedded programming concepts (Arduino/ESP32 in C++, ROS 2 in Python) for controlling sensors, actuators, and robot behavior.	C3	GA 3	Class Activity, Quiz, Assignments, Mid Exam, Final Exam
C3	Design and implement algorithms to solve real-world robotics problems, such as obstacle avoidance, sensor data processing, and differential-drive robot navigation.	C3	GA4	Worksheets, Project

Domains:

C=Cognitive, A=Affective, P=Psychomotor

Levels:

Cognitive = {1: Remembering, 2: Understanding, 3: Applying, 4: Analyzing, 5: Evaluating, 5: Creating}

Affective = {1: Receiving, 2: Responding, 3: Valuing, 4: Organizing, 5: Characterizing}

Psychomotor= {1: Imitation, 2: Manipulation, 3: Precision, 4: Articulation, 5: Naturalization}



Aror University of Art, Architecture, Design & Heritage Sukkur.

Course Outlines:

Week	Lec#	Topic	Completion %
1	1	Introduction to Robotics — definitions, history, evolution of robotics, applications across industries	3%
	2	Robot Architecture — types of robots, sensors/actuators/controllers overview, open vs. closed-loop control systems	6%
2	1	Microcontrollers Overview I — Arduino boards (Uno, Mega), Raspberry Pi architecture and use cases	9%
	2	Microcontrollers Overview II — ESP32 and Raspberry Pi Pico; digital vs. analog I/O concepts, pin mapping	13%
3	1	Sensors — classification, examples (touch, light, temperature), interfacing basics with Arduino	16%
	2	Serial Communication — UART basics, Serial Monitor, debugging techniques	19%
4	1	Analog Inputs — ADC concepts, resolution, sampling, reading analog sensors	22%
	2	PWM Control — pulse width modulation concept, duty cycle, analog output emulation	25%
5	1	Sound Devices — buzzers (active/passive), piezoelectric principles, tone generation	28%
	2	Sound Devices — applications, alert/feedback system design, sound-based signaling in robotics	31%
6	1	Obstacle Sensors — ultrasonic sensor principles (HC-SR04), distance measurement, NewPing library	34%
	2	Obstacle Sensors — IR sensors, ultrasonic vs. IR comparison, obstacle detection strategies	38%
7	1	Actuators — DC motors, motor drivers/controllers (L298N/H-bridge)	41%
	2	Actuators — servo motors and stepper motors, control principles and applications	44%
8	1	Integration — combining sensors and actuators, system design considerations	47%
	2	Integration — project planning, prototyping approach, troubleshooting strategies	50%
9	1	ESP32 Introduction — setup, pinout, architecture, programming environment	53%
	2	ESP32 Wi-Fi — Wi-Fi basics, web server concepts, client-server communication	56%
10	1	IoT Concepts — HTTP protocol basics, RESTful communication	59%



Aror University of Art, Architecture, Design & Heritage Sukkur.

	2	MQTT and Cloud Dashboards — MQTT protocol, ThingSpeak/Streamlit dashboard concepts	63%
11	1	Introduction to ROS — architecture, ROS ecosystem, ROS1 vs. ROS2 overview	66%
	2	ROS Nodes, Topics, and Messages — publisher–subscriber model, message types	69%
12	1	Degrees of Freedom and Coordinate Frames — DOF concept, reference frames, frame transformations	72%
	2	URDF — links, joints, visual/collision/inertial tags, robot description structure	75%
13	1	Forward Kinematics and Differential Drive Kinematics — FK using DDBot as example; wheel velocity to (v, ω) relationship	78%
	2	TF (Transform Tree) — tf2 concepts, frame trees, odom $\rightarrow$ base link $\rightarrow$ wheel frame relationships	81%
14	1	Introduction to Gazebo — simulation basics, ignition, physics engine overview	84%
	2	Gazebo Simulation Environment — world files, spawning robots, topics in simulation	88%
15	1	DDBot in Gazebo — SDF format, robot description for simulation	91%
	2	DDBot in Gazebo — plugins, nodes, sensor/actuator simulation	94%
16	1	Bridging ROS and Gazebo — ROS–Ignition bridge concepts, message bridging	97%
	2	Final Project Briefing — project guidelines, evaluation criteria, integration expectations	100%

Text Books:

- Introduction to Robotics: Mechanics and Control by John J. Craig
- A Concise Introduction to Robot Programming with ROS2" by Francisco Martín Rico

Reference Books:

Books:

- A Concise Introduction to Robot Programming with ROS2
- Arduino Robotics" by John-David Warren, Josh Adams, and Harald Molle
- ~~"Robotics, Vision and Control: Fundamental Algorithms"~~ by Peter Corke

Prepared by _____

Verified by _____



Aror University of Art, Architecture, Design & Heritage Sukkur.

HOD _____



Aror University of Art, Architecture, Design & Heritage Sukkur.
